

AP

FGD SYSTEMS CAN REDUCE SO_2 EMISSIONS FROM THERMAL PLANTS BY 50-90%, BUT COSTS AND SHORT SUPPLY REMAIN DETERRENTS

country to meet the nation's burgeoning electricity requirements. This even as India reiterated its commitment at the ongoing COP28, the annual global climate change meet, in Dubai, to ensure that 50 per cent of its power generation capacity in 2032 will come from non-fossil fuel sources.

At the same time, the Centre will have to ensure that it lessens the amount of pollution being emitted from coal-fired thermal plants. Coal combustion emits large amounts of sulphur dioxide and nitrogen oxide, which make up the toxic particulate matter that hangs in the atmosphere as poisonous smog causing respiratory trouble for humans. The problem is fairly acute in Delhi, going by an estimate the Centre for Science and Environment (CSE) presented on November 27, according to which thermal power plants contribute around 8 per cent to the PM in the air in Delhi-NCR. Coal-based thermal power plants were held responsible for aggravating the air pollution in Delhi-NCR, since a majority do not comply with emission control norms. This despite the MoEFCC notifying emissions standards for thermal power plants for SOx and NOx, and revising norms for PM in 2015. The notification required compliance within two years, but a five-year extension was given later to all except those in Delhi-NCR, which were given until 2019 to comply due to the region's high pollution levels.

CSE's study revealed that only two—the one at Dadri and the Mahatma Gandhi plant in Jhajjar—out of the 11 thermal plants in the vicinity have the SO_x control measures in place, and only the Jhajjar plant meets the emissions standards for SO_x. To control SO_2 emissions, thermal plants are required to install flue gas desulfurisation (FGD) systems, but apart from the additional cost, complex installation, the thermal plants find it difficult to source the equipment. Chinese manufacturers dominate the FGD equipment making, but post the Galwan skirmish, shipments with FGD equipment are caught in the diplomatic turmoil.

The officials at the power ministry's watchdog—the Central Electricity Authority or CEA—confirm that despite extensions, a bulk of the installed capacities, not only in the NCR but across the country, will fail to adhere to the deadlines. The environment ministry graded thermal power plants into three categories and gave them time depending on their proximity to the big cities. So, Category A included plants within a 10-km radius of Delhi-NCR or cities with a million-plus population; Category B included plants within a 10-km radius of critically polluted areas; while the rest fell in Category C. They were given extended deadlines of 2024, 2025

SOLUTIONS

- **Centre notified emissions standards in 2015 for thermal power plants for SO_x and NO_x, and revised PM norms**
- **Smoothen out imports of flue gas desulfurisation (FGD) systems used to control SO_2 emissions in thermal plants**
- **Use advanced power generation technologies. For instance, an advanced firing system using pure oxygen instead of air eliminates thermal NO_x production**